Mercy House

Can a detective prove what happened when a plausible story is wrong?

> A snowbound town in 1900. Local memories. Physical traces. No answer checklist.

Eight tasks · Four case families · Three designed levels.

Games & World Simulations / Open-compatible. Judges bring the detective; our deliverable is the environment.

A real progression

Case family	Challenge
Lantern Room / beginner	Witnessed fatal assault; learn tools and sourced reporting
Second Passenger / intermediate	Two deaths; identify sabotage and intended target
Bell Before Dawn / advanced	Forged foreknowledge, delivery chain and an unrelated-looking death
Mercy House / five scopes	Loyalty, concealed evidence, fabricated threats and historical crimes

New cases change the assailant with the seed and regenerate actions, traces and recollections. They are original authored scenarios using broad mystery mechanisms.

Causes execute before arrival

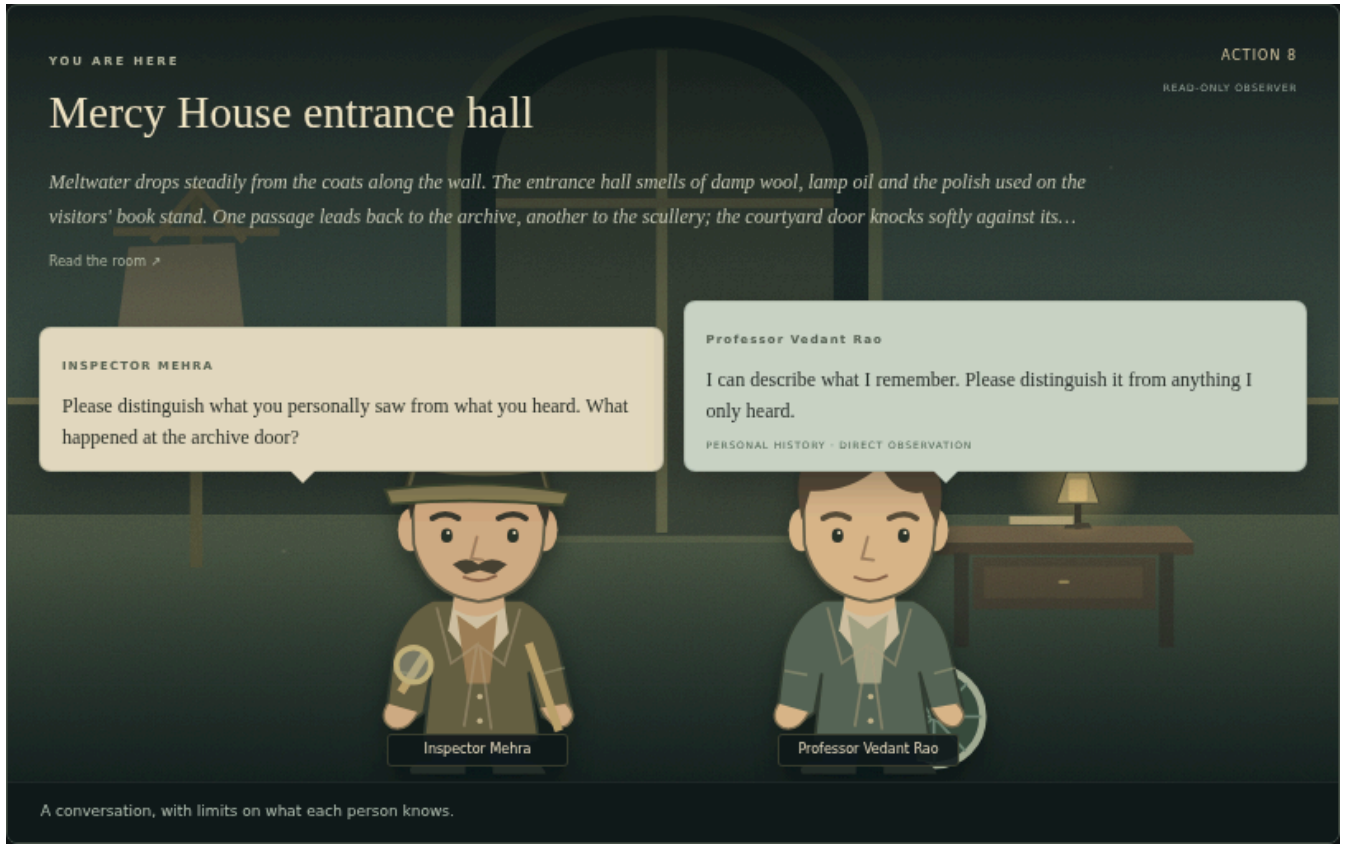
Validated actions → local perceptions → persistent evidence → investigation

Remove the assault: no body. Leave the brake intact: the coach survives. Leave the cordial unadulterated: drinking does not kill.

Witnesses know what they saw, not an offstage cause of death.

Canonical tape: recorded LLM decisions. New families: authored seeded tapes. Neither is claimed to be a continuously autonomous town.

Watch the actual inquiry



Places change after movement.

Dialogue and evidence follow public observations.

Provider setup, live streaming and labeled replays.

No hidden character memories or invented thoughts.

Shown: an actual scripted-reference replay. The UI labels it explicitly. New live runs still use strict scoring.

Reward proof, not a lucky name

```
R = clamp(C − 0.20 F − 0.10 U, 0, 1)
```

C: supported target coverage. F: distinct false findings. U: distinct unsupported findings.

Physical mechanisms and testimony must agree. Paper ownership alone cannot establish forgery. An unsupported correct guess earns zero.

Narrative is retained but unscored. Programmatic authored proof paths; no LLM judge.

Protect the answer from the solver

Role	Authority
Shell workspace	Notes, scripts and validated public town tools
Trusted harness	Budget, parsing, public logging and orchestration
Protected world	Hidden state, transitions, provenance and reward
Model gateway	Credentials and fixed endpoint configuration

Private planner → gated recollections → public speaker. Actor questions cannot directly inspect private memories.

Measured beginner success

4 / 4 live detective runs · reward 1.0

Seeds	NPC condition / conclusion
211, 307, 401	Native API detective and live isolated character agents
37	Additional opposite-assailant calibration

No false or unsupported findings. The actual acquired proof and raw runs are retained.

Small development sample, one model deployment. Not a reliability estimate. No complete live success established for the other families.

RL: useful evidence, honest failure

Evaluation	Result
20,000 CPU training episodes	530.243 seconds; 460 coefficients; no GPU
Historical offline greedy checkpoints	147/160 versus untrained 1/160
Historical live checkpoint smoke	2/5; not whole cases
New matched family comparison	LLM-ranked 1/3; frozen selector 0/3

Same 2–6 ranked candidate contract, fixed grounded response bank and normal disclosure gates. Three cases only; branch-dependent proposals differ. Frozen policy showed no whole-case advantage and remains optional.

Evidence under stress

Check	Result
Eight task reference/no-op pairs	1.0 / 0.0 for every task
Town + legacy + Node tests	96 + 9 + 4 pass
Real-command isolation	Private routes denied; workspace forgery scores zero
Causal counterfactuals	No destructive action → no corresponding death
No-cache four-image build	14.069 seconds locally

Failed runs stay in the archive. Scripted references are not presented as learned or LLM results.

What this entry establishes

A reproducible investigation environment with an accessible demo and evidence-grounded difficulty.

Local tools, isolated character memories, strict scoring, observable failures and measured CPU RL.

Limits: authored proof vocabulary, small samples, no established whole-case RL gain, incomplete offstage-life and supernatural mechanics.

Final separate character-model deployment and organizer-machine timing must be verified. User handles the private repository and submission form.